

4EB00 Special Topic Thermodynamics and Combustion

Cycle analysis of a jet-engine
Lecture 1

Bart Somers (P&F/ICE)



Technische Universiteit
Eindhoven
University of Technology

Where innovation starts

Primary energy demand per sector

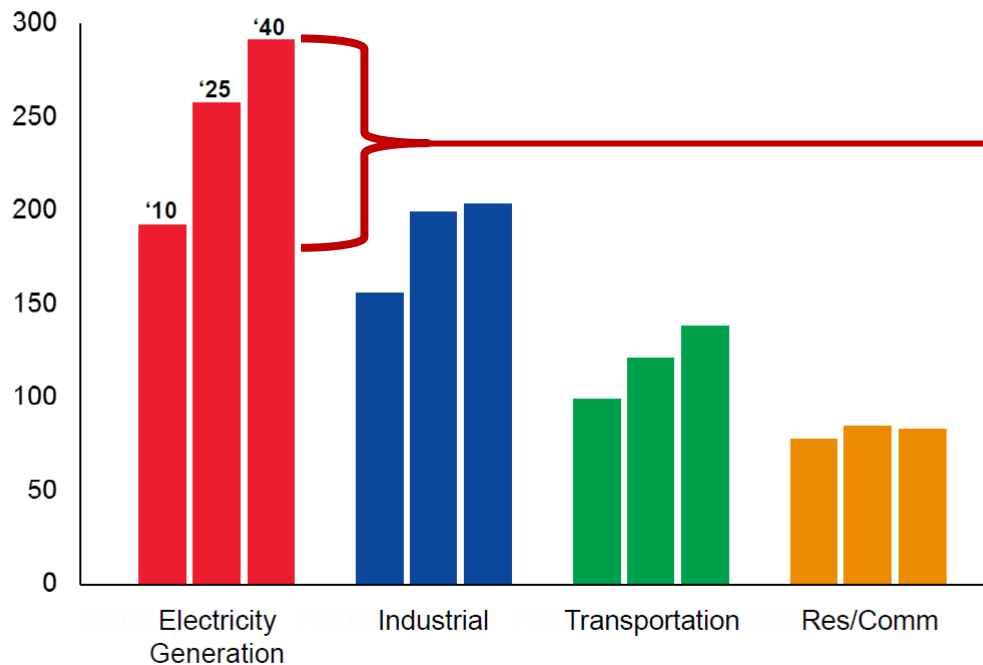
Quadrillion=10¹⁵.
10¹⁵ J = Peta Joule

1000 BTU~1MJ

Electricity Generation Leads Growth

Primary Energy Demand by Sector

Quadrillion BTUs



Electrification of
leads to
increase in
energy demand

ExxonMobil

ExxonMobil 2015 Outlook for Energy

Electrification is the main driver

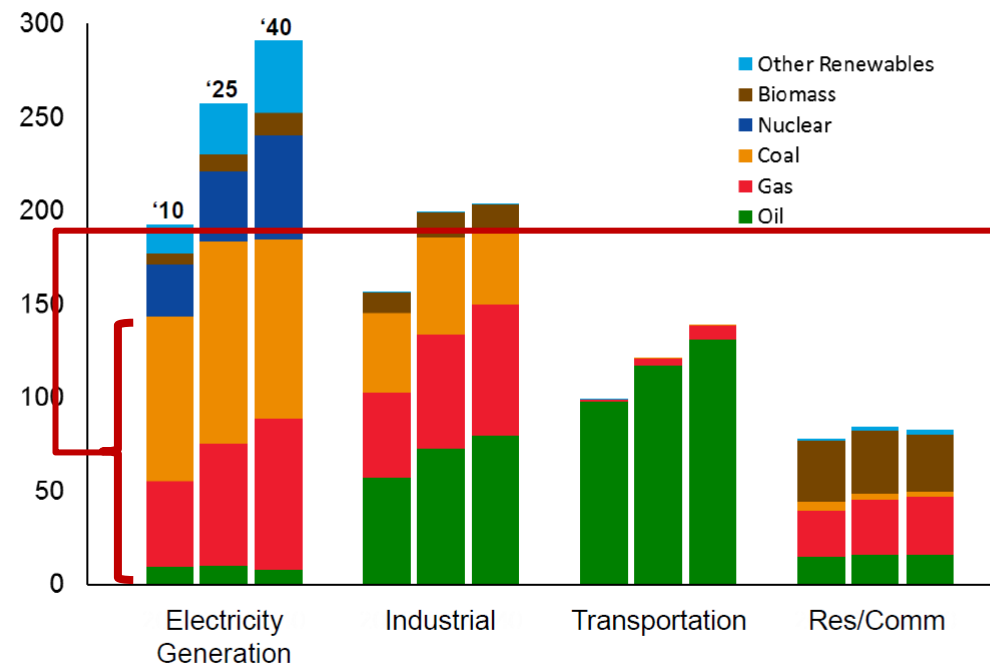
Quadrillion = 10^{15} .
 10^{15} J = Peta Joule

1000 BTU ~ 1MJ

Electricity Generation Leads Growth

Primary Energy Demand by Sector

Quadrillion BTUs



Need for fast
increase in
Renewable
Electricity to
reduce Carbon
footprint!!

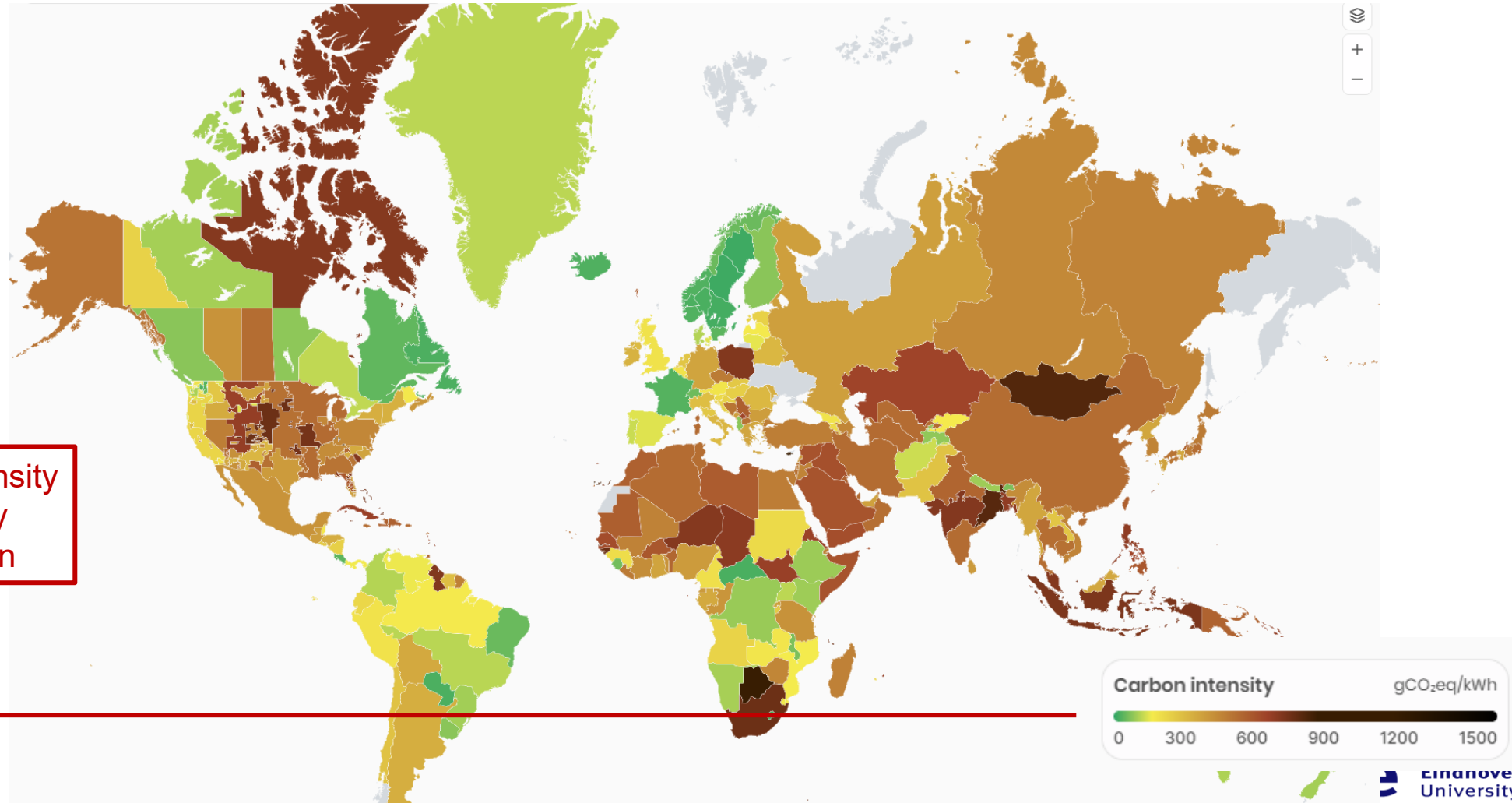
ExxonMobil

ExxonMobil 2015 Outlook for Energy

Carbon footprint of electricity is a local property

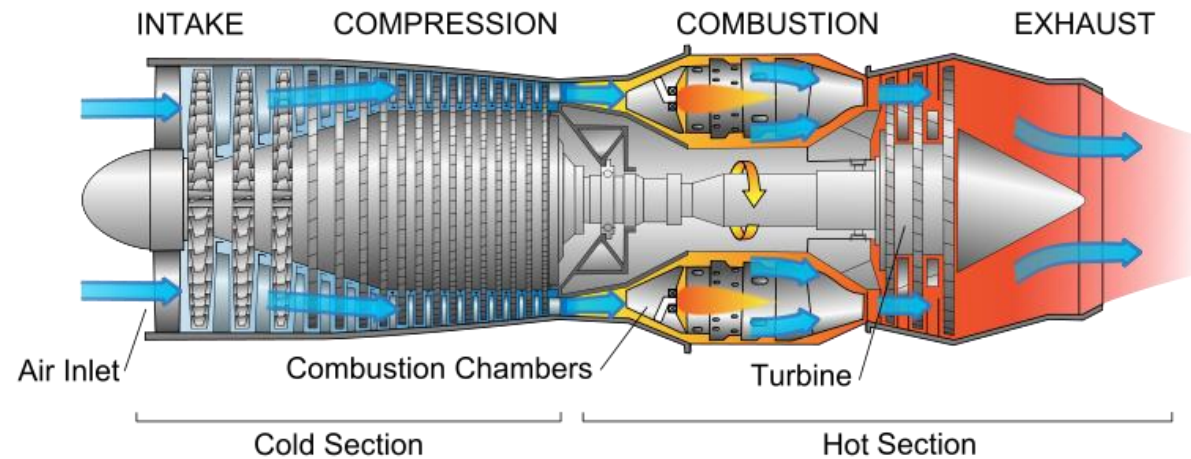
<https://app.electricitymaps.com/>

2025



The JetEngine

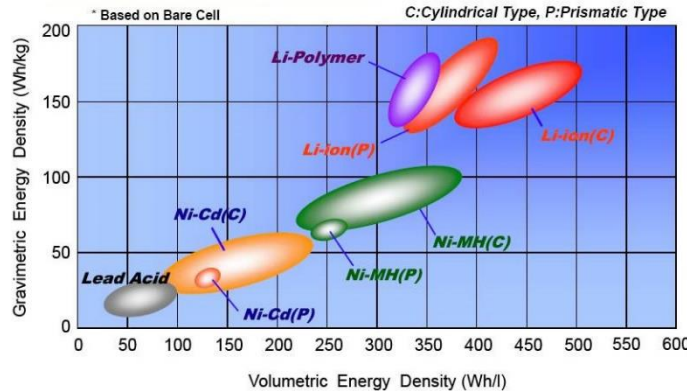
- **Combustion is key: the combustor drives the application!**



- **A member of the 'hard-to-electrify modalities'**
 - **Planes**
 - **Shipping**
 - **Long Haul Transport**

Hard to electrify: reason 1 - energy density

Beercoaster computation



Average weight kerosene: 0.78-0.82 kg/liter
150,000 liters = 120 tonnes
Boeing 747 10hr flight

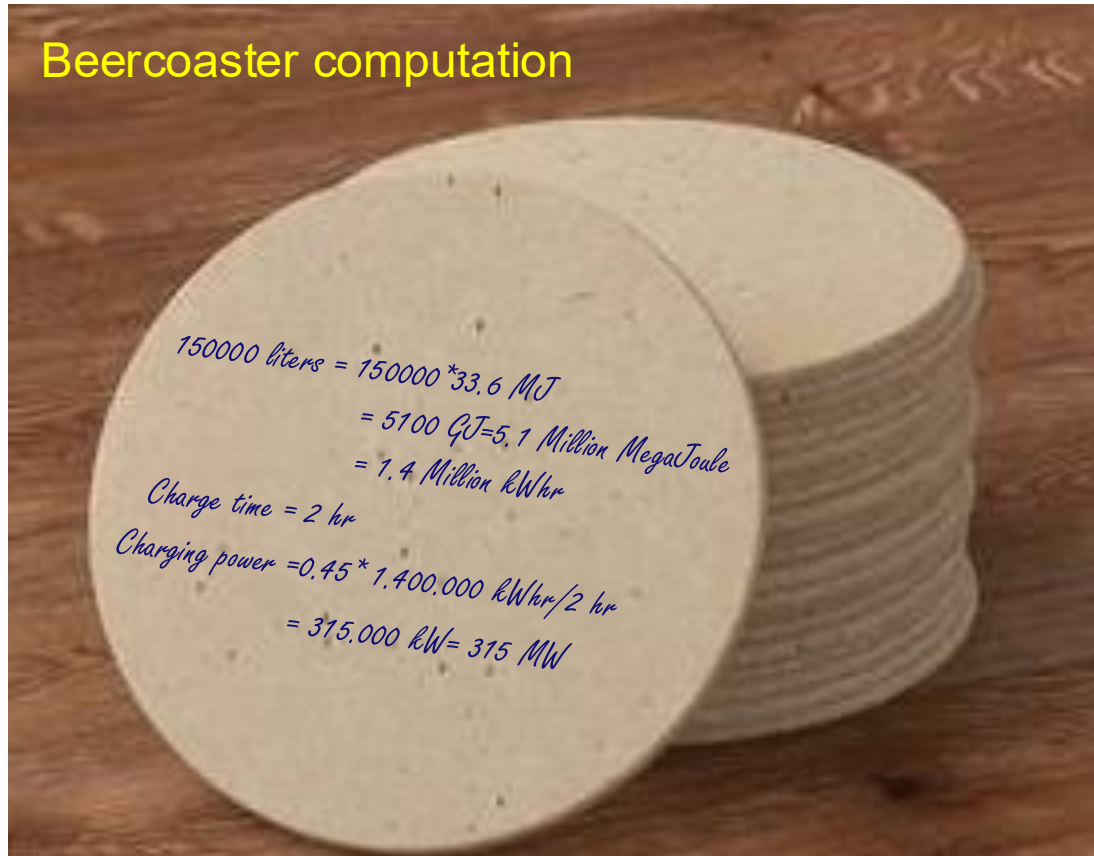
Average weight Li-ion battery: 5-7 kg per kWhr
100 kWhr* = 100*3.6 MJ = 0.36 GJ
500 - 700 kg

*1kWhr = 1000*3600 J = 3.6MJ

$150000 \text{ liters} = 150000 * 33.6 \text{ MJ}$
 $= 5100 \text{ GJ}$
 $= 1.400.000 \text{ kWhr}$
 $0.45 * 1.400.000 \text{ kWhr} * 5 = 3600 \text{ tonnes}$
Max Payload 747 = 112 tonnes
MTOW 747 = 336 tonnes
MLW 747 = 225 tonnes

Hard to electrify: reason 2 - charging infra

Beercoaster computation



Schiphol has 940 flights
a day = 77 per 2hr

$$= 77 * 315 \text{ MW} = 24 \text{ GW}$$



Largest Dutch Power plants

- Claus powerplant: 1.9 GW

Largest offshore wind farm

- Hollandse Kust Zuid: 1.5 GW (Dutch)
- Dogger Bank: 3.6 GW (UK)

Contents of the Lecture

- **Ideal Gas mixtures** (*combustion is about changes in mixture comp*)
 - **Thermodynamic Properties of a mixture (2.9)**
 - Define a mixture
 - Equation of State
 - Calorific properties like heat capacity, energy and enthalpy
 - **Explosion temperature (and pressure) of a tank, an application of Nasa thermodynamic tables.**
- **Cycle analysis of a jet engine using thermodynamic tables**
 - Combustor
 - Diffusor
 - Etc..

Ideal gas mixtures (definition)

- How to define/characterize a mixture (*J number of components*)
- Mole fractions (*N_i – number of moles of component i*)

$$X_i = \frac{N_i}{\sum_{i=1}^J N_i} = \frac{N_i}{N_{tot}}$$

- Mass fractions (*M_i – mass of component i*) = $M_i = N_i \mu_i$

$$Y_i = \frac{M_i}{\sum_{i=1}^J M_i} = \frac{M_i}{M_{tot}}$$

Ideal gas mixtures (Equation of State)

- Equation of state (double ideality)
 - Each species ideal gas (ideal solution)

$$P_i V = N_i R_u T$$

P_i = so-called Partial Pressure of component i

- Sum it over all constituents

$$\sum_{i=1}^J P_i V = \sum_{i=1}^J N_i R_u T$$

$$= N_{tot} R_u T = PV$$

Dalton's law

$$\sum_{i=1}^J P_i = P$$

Double
ideality

Because mixture is also
considered to be an ideal gas

Some words on heat capacity and Temperature

What is Temperature of a gas

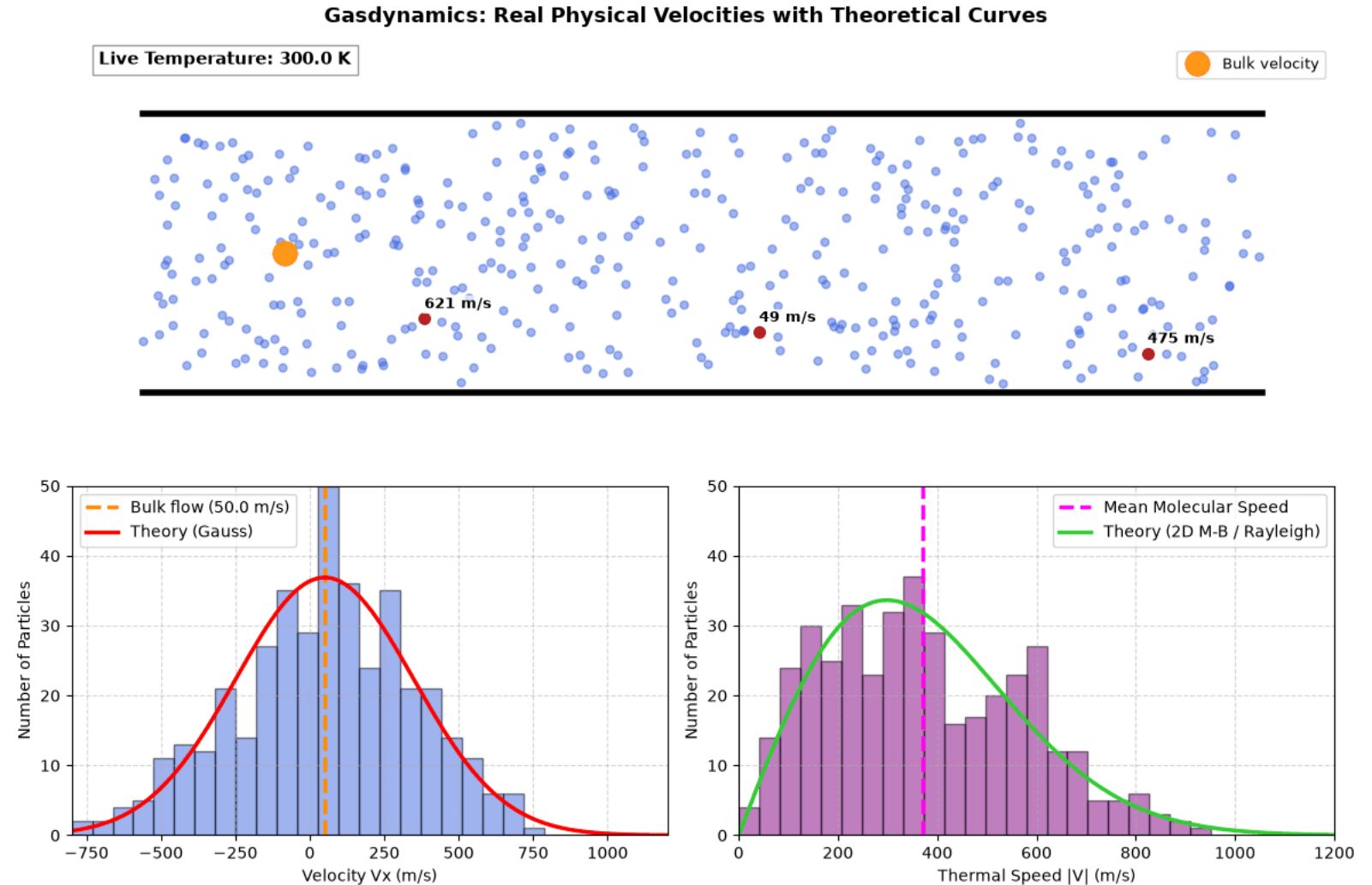
- Temperature ‘measures’/’is a measure of’ the average kinetic energy of molecules
- Example:
 - average flow is 50 m/s (●)
 - Temperature increases (●) at certain times

*
Asked AI to generate a Python script for this!

Some words on heat capacity and Temperature

T increases at certain times

- Rayleigh distribution instead of MB (2D vs 3D)
- Bulk velocity is 50 m/s (> Windkracht 11, Beaufort scale 11)

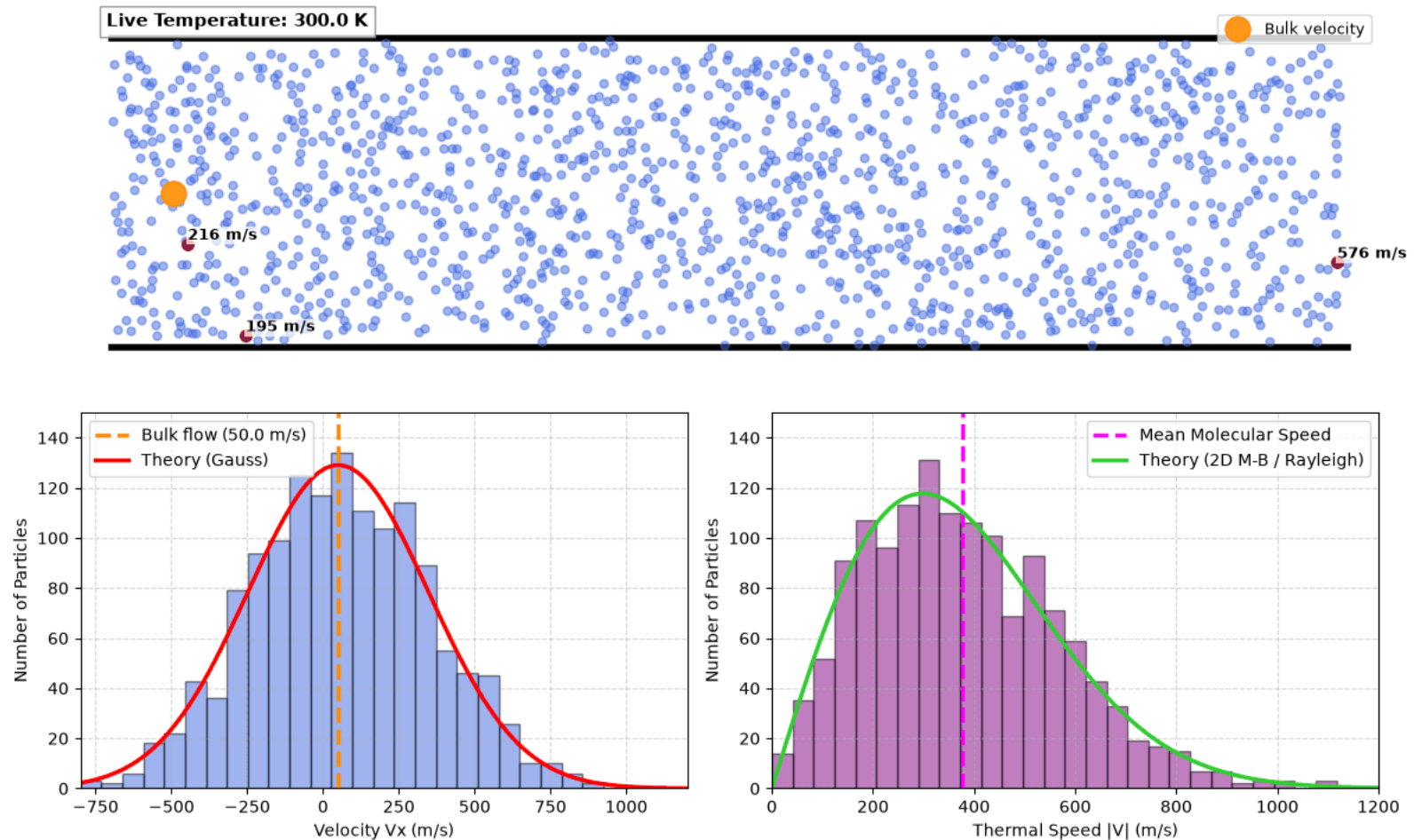


Some words on heat capacity and Temperature

T increases at certain times

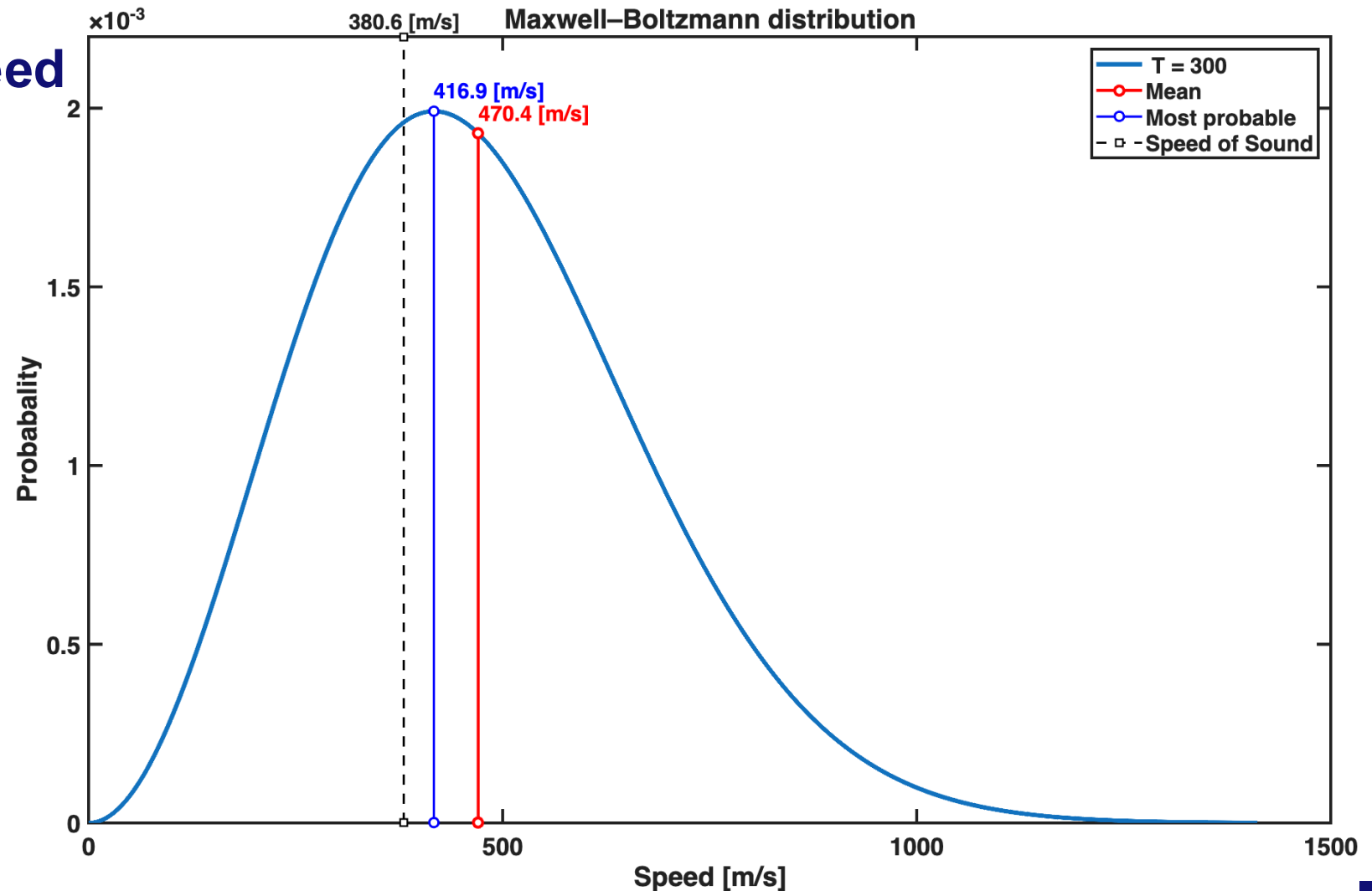
- Rayleigh distribution instead of MB (2D vs 3D)
- Bulk velocity is 50 m/s (> Windkracht 11, Beaufort scale 11)

Gasdynamics: Real Inter-Particle Collisions (Hard-Sphere)



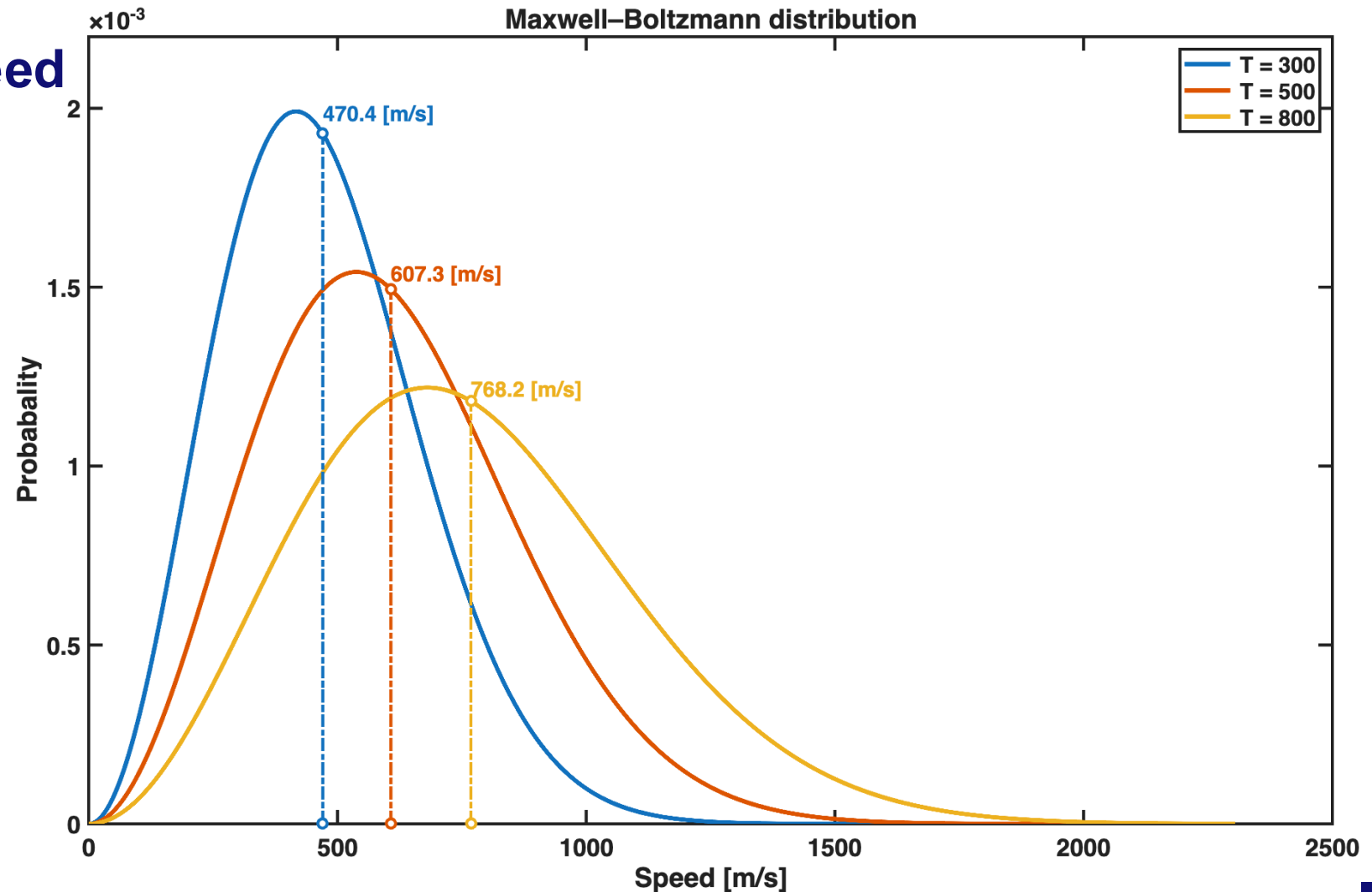
Some words on heat capacity and Temperature

Maxwell-Boltzmann speed distribution



Some words on heat capacity and Temperature

Maxwell-Boltzmann speed distribution



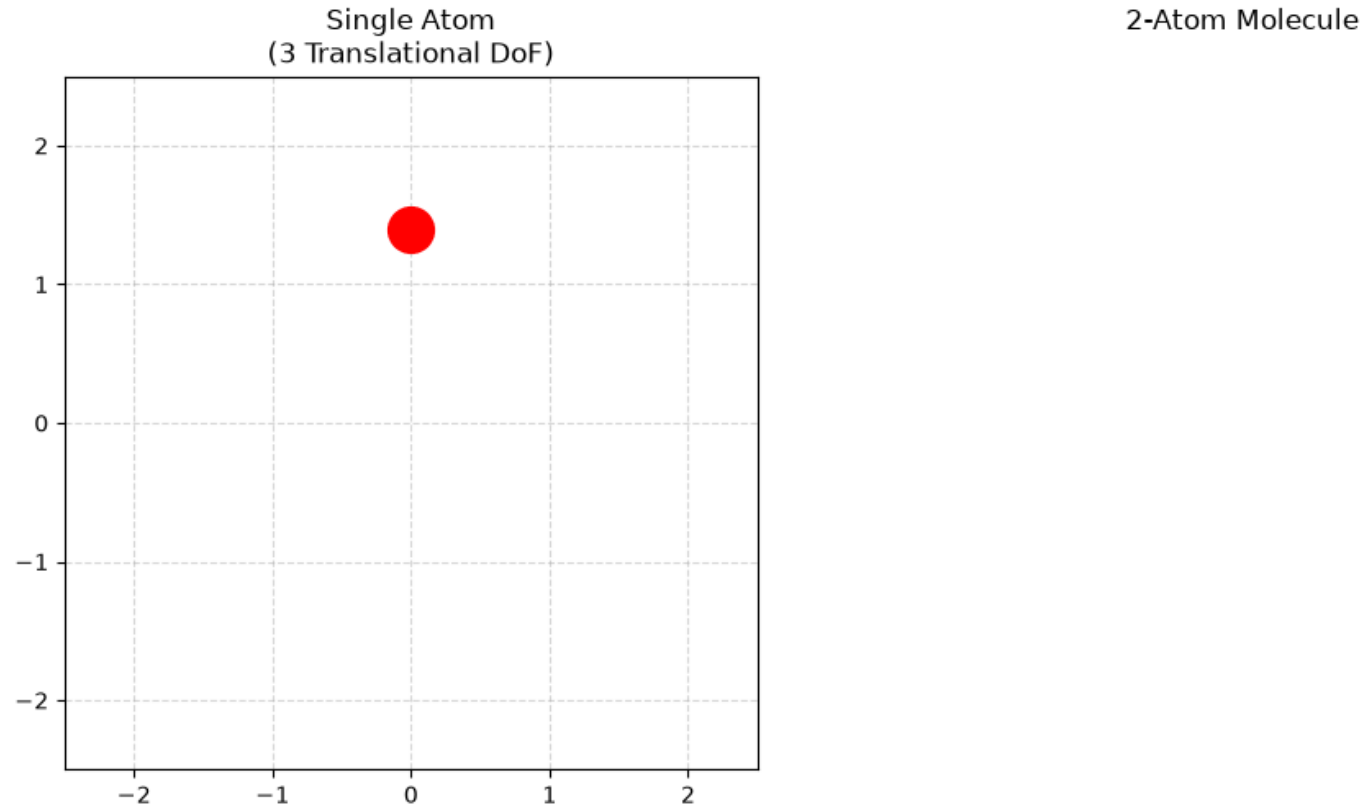
What about heat capacity $c_V = \frac{de}{dT}$

Single atom molecule

$$e = e_{\text{kin}} = \frac{3}{2} k_B T$$

$$c_V = \frac{de}{dT} = \frac{3}{2} k_B$$

Molecular Degrees of Freedom



What about heat capacity $c_V = \frac{de}{dT}$

Single atom molecule

$$e = e_{\text{kin}} = \frac{3}{2} k_B T$$

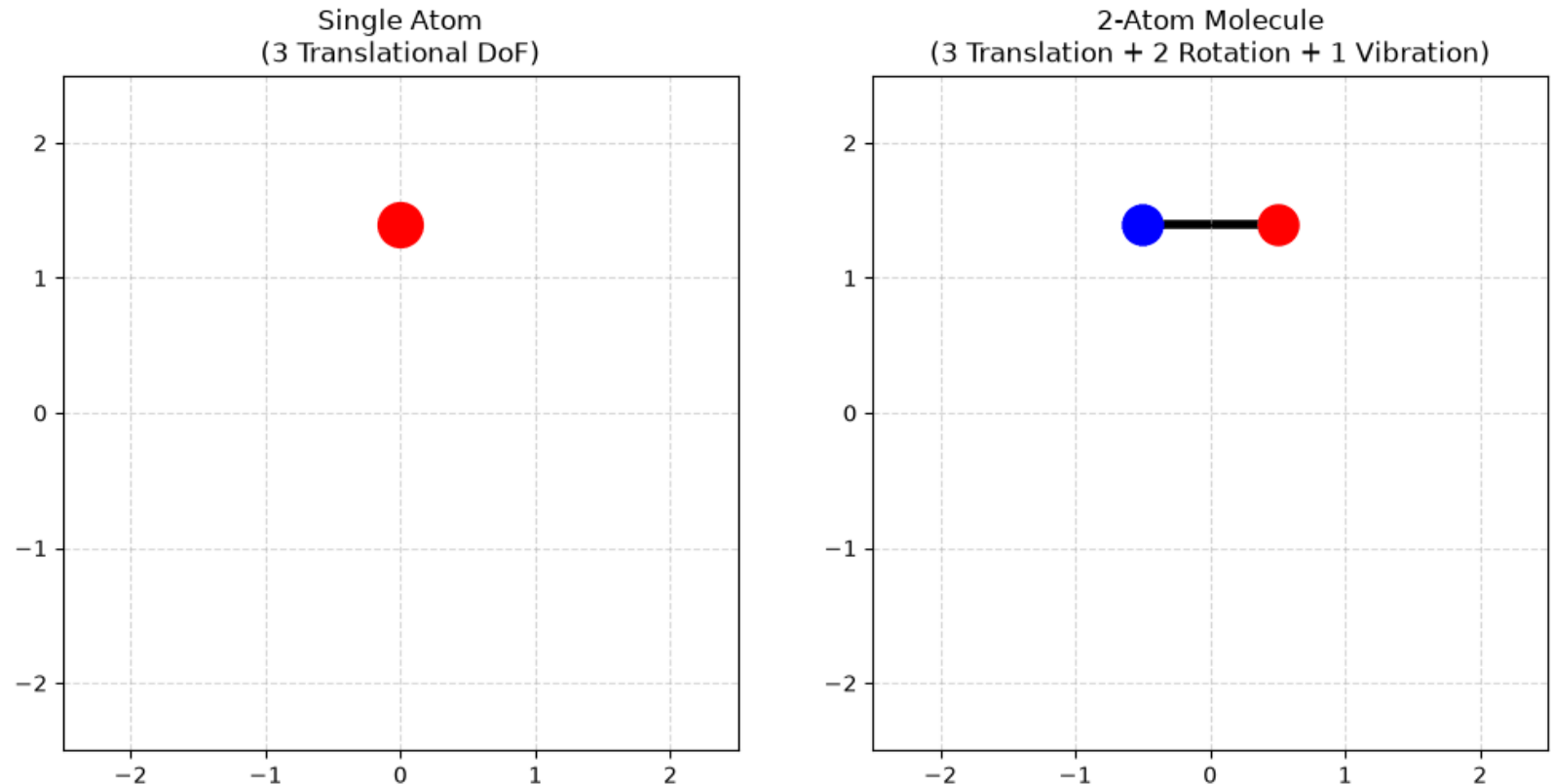
$$c_V = \frac{de}{dT} = \frac{3}{2} k_B$$

Two atom molecule

$$e = e_{\text{kin}}^3 + e_{\text{rot}}^2 + e_{\text{vib}}^1 =$$

$$c_V = \frac{de}{dT} = \frac{6}{2} k_B$$

Molecular Degrees of Freedom



NASA polynomials

$$c_V = \frac{de}{dT}$$

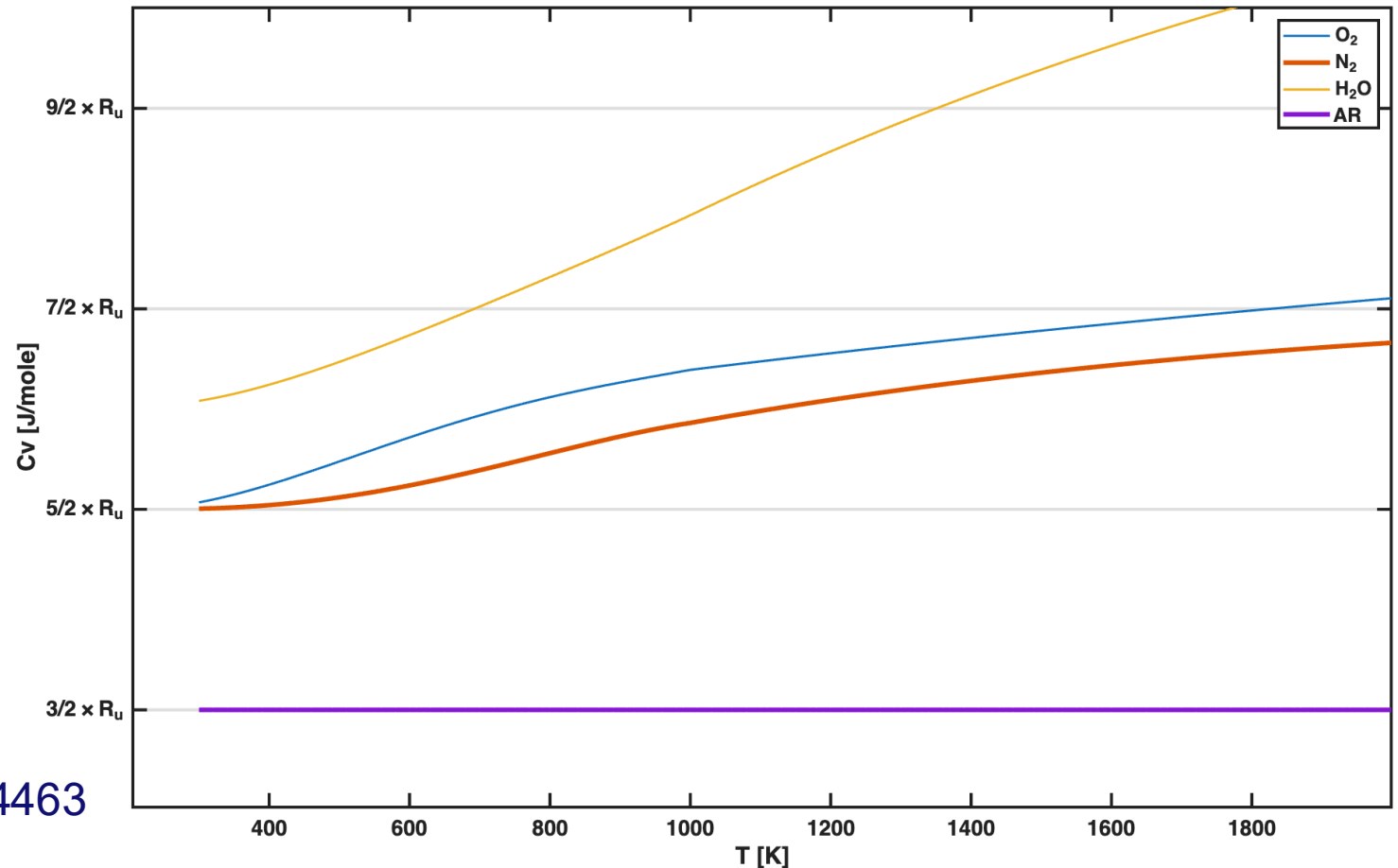
One mole of Single atom molecules

$$c_V = N_A \frac{3}{2} k_B = \frac{3}{2} R_u$$

One mole of Two atom molecules

$$c_V = N_A \frac{6}{2} k_B = \frac{6}{2} R_u$$

$$\left. \begin{array}{l} k_B = 1.380649 \cdot 10^{-23} \\ N_A = 6.0221408 \cdot 10^{+23} \end{array} \right\} R_u = 8.314463$$



Standardized properties: NASA *polynomials*

$$h_i(T) = h_{f,i}^0 + \int_{T_{ref}}^T c_p(T') dT' \text{ for each component } i \text{ in the mix}$$

For determining conditions during re-entry of the Apollo return pod NASA has determined these standardized properties for many components.

Makes life easy for us. Let's use them.
Exercises and their answers can be used to check your approach in the assignment.



Standardized Properties (consistent)

Standardized properties for each component i

$$h_i(T) = h_{f,i}^0 + \int_{T_{ref}}^T c_p(T') dT' \text{ for each component } i \text{ in the mix}$$

is the so-called 'formation enthalpy'

$$h_{f,i}^0$$

The energy it costs to create a molecule from its basic elements (CHON)

Standardized properties: NASA polynomials

Example HNasa.m

```
1 function [H,HU] = HNasa(T,Sp)
2 %HNasa(T,Sp):: computes internal specific enthalpy of species at Temp T
3 %
4 %   Input: T, temperature
5 %           Sp, thermal database entry. So-called NASA polynomials
6 global Runiv
7 H=zeros(size(T));HU=H;
8 if (isempty(Runiv))
9     fprintf('[HNasa] Assign global Runiv\n');
10    return
11 end
12 if (isnan(Sp.Ts))
13     Tl=T;
14     a=Sp.Pol(1,:);
15     HU=a(6) + a(1).*Tl+(a(2)/2).*Tl.^2+(a(3)/3).*Tl.^3+(a(4)/4).*Tl.^4+(a(5)/5).*Tl.^5; % Formula 5.4 of lecture notes
16 else
17     ilow = (T <= Sp.Ts);
18     Tl=T(ilow);
19     a=Sp.Pol(1,:);
20     HU(ilow)=a(6) + a(1).*Tl+(a(2)/2).*Tl.^2+(a(3)/3).*Tl.^3+(a(4)/4).*Tl.^4+(a(5)/5).*Tl.^5;
21     5h order polynomial in T
22     ihigh = (T > Sp.Ts);
23     Tl=T(ihigh);
24     a=Sp.Pol(2,:);
25     HU(ihigh)=a(6) + a(1).*Tl+(a(2)/2).*Tl.^2+(a(3)/3).*Tl.^3+(a(4)/4).*Tl.^4+(a(5)/5).*Tl.^5; % Formula 5.4 of lecture notes
26 end
27 H=HU.*Runiv/Sp.Mass;
28 end
```

Standardized properties: NASA polynomials

Exercise1Start.m

```
clear all;close all;clc;

%%
warning off
abspath_to_generalfolder='D:\users\lsomers\Documents\1 Lectures\4EB00\2015\Matlab\General'; % absolute reference to General folder
addpath(abspath_to_generalfolder); % assumes it is in the current directory.

%% Some easy units
kJ=1e3;kmol=1e3;

%% Used by Nasa pols
global Runiv pref
Runiv=8.314472;
pref=1.01235e5; % Reference pressure, 1 atm!
Tref=298.15; % Reference Temperature

%% Load Nasadatabase
TdataBase=fullfile('General','NasaThermalDatabase');
load(TdataBase);
whos
% Nasa is ready

%%
T=[300:10:4000]; % Define T vector
iSp1=myfind({Sp.Name},{ 'O2' }); % Find index for O2
iSp2=myfind({Sp.Name},{ 'O' }); % Find index for O

hO2=HNasa(T,Sp(iSp1)); % Compute specific enthalpy for O2 as function of T vector
MO2=Sp(iSp1).Mass; % Molar mass (kg/mol)
hO=HNasa(T,Sp(iSp2)); % Compute specific enthalpy for O as function of T vector
MO=Sp(iSp2).Mass; % Molar mass (kg/mol)
```

Note: The functions do **NOT EXTRACT** a value from the database. They **COMPUTE THE VALUE** using the polynomial parameters you provide via the Sp-struct from the database and the T you provide to the function.

Standardized properties: Nasa polynomials

At home exercise 1: Compute h of O_2 and O at T_{ref} (25 degC)

1. determine $h_{O_2}(T_{ref})$
2. determine $h_O(T_{ref})$, see also book page 130

3. Determine enthalpy of formation

$$h_{f,O}^0, h_{f,N}^0, h_{f,H}^0$$

Standardized properties: Nasa polynomials

Answers exercise 1:

1. determine $h_{O_2}(T_{ref}) = 5.11e-04 \text{ [J/kg/K]}$
2. determine $h_O(T_{ref}) = 1.56e+07 \text{ [J/kg/K]}$ see also book page 130

3. Enthalpy of formation

$$\begin{aligned}h_{f,O}^0 &= 1.56e+07 \text{ [J/kg/K]} \\h_{f,N}^0 &= 3.37e+07 \text{ [J/kg/K]} \\h_{f,H}^0 &= 2.16e+08 \text{ [J/kg/K]}\end{aligned}$$

Ideal gas mixtures

Calorific relations for a mixture

- Nb u is e (energy), h is enthalpy

- Per unit mass

$$u_{mix}(T) = \sum_i Y_i u_i(T)$$

$$c_{V,mix}(T) = \sum_i Y_i c_{V,i}(T)$$

etc...

Ideal gas mixtures

- How to define/characterize a mixture

- Mole fractions

$$X_i = \frac{N_i}{\sum_{i=1}^J N_i} = \frac{N_i}{N_{tot}}$$

- Mass fractions

$$Y_i = \frac{M_i}{\sum_{i=1}^J M_i} = \frac{M_i}{M_{tot}}$$

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Ideal gas mixtures

Exercise 2: Compute properties of air.

1. Determine composition of air(N₂/O₂)
2. Determine
3. Determine specific heat capacity

X_i, Y_i
 $h_{air}(T_{ref})$. And plot $h_{air}(T = [250 - 2000])$

$c_{p,air}(T_{ref}), c_{p,air}(T = [250 - 2000])$

Ideal gas mixtures

Exercise 2: Compute properties of air.

1. Determine composition of air(N2/O2)
2. Determine
3. Determine specific heat capacity

$$X_i, Y_i$$

$$h_{air}(T_{ref}), h_{air}(T = [300 - 2000])$$

$$c_{p,air}(T_{ref}), c_{p,air}(T = [300 - 2000])$$

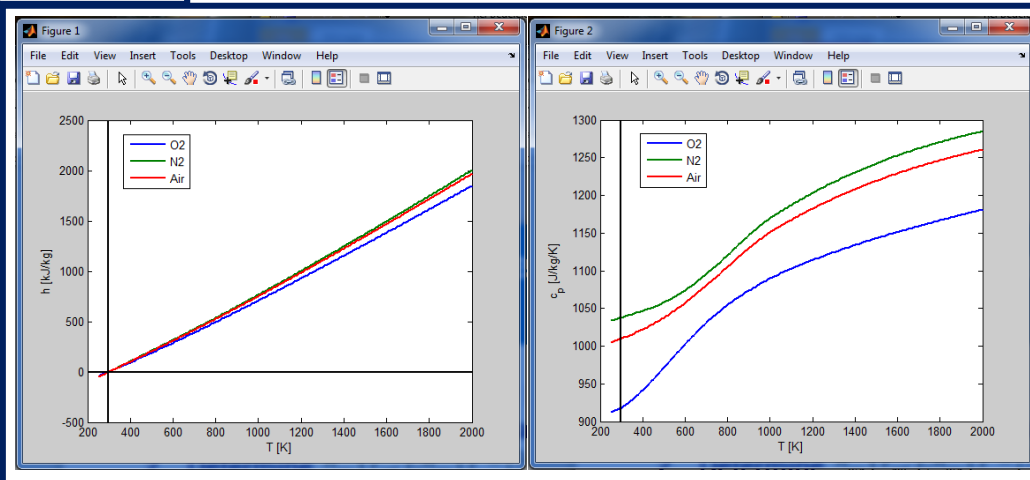
Ans1

Name	Xi	Yi
O2	0.210000	0.232918
N2	0.790000	0.767082

Ans2/3

Name	href	cpref
O2	0.00051	918.104
N2	51.04356	1037.755
Air	39.15474	1009.886

Ans2/3



Contents

- **Ideal Gas mixtures**
 - Thermodynamic Properties of a mixture (2.9)
 - **Exercises: Constant volume explosion (ex3), Adiabatic flame temperature (ex4)**
(application of Nasa thermodynamic tables).
- **Cycle analysis of a jet engine using thermodynamic tables**
 - Combustor
 - Diffusor
 - Etc..

Ex3 Constant Volume Explosion

Model System



- Adiabatic
- Constant volume
- Closed
- Constant mass

$$E_2 - E_1 = \cancel{Q} - \cancel{W} + \cancel{\sum_{boundaries} \dot{m}_b h_b}$$

$$E_2 - E_1 = 0$$

$$u_{mix,2} - u_{mix,1} = 0$$

Ex3 Constant Volume Explosion

- Apparently

$$u_{mix,2} - u_{mix,1} = 0$$

- Given initial state

- 0.15 moles of acetylene (C_2H_2) and 2 moles of air. Volume of vessel is 50 liter. Due to fire T=800K. Fire is out. Mixture is going to ignite.



Ideal gas mixtures

- Calorific relations for a mixture

- Per unit mass

$$u_{mix}(T) = \sum_i Y_i u_i(T)$$

$$c_{v,mix}(T) = \sum_i Y_i c_{v,i}(T)$$

etc...

- Per unit mole

$$\bar{u}_{mix}(T) = \sum_i X_i \bar{u}_i(T)$$

$$\bar{c}_{v,mix}(T) = \sum_i X_i \bar{c}_{v,i}(T)$$

etc...

Ideal gas mixtures

- How to define/characterize a mixture
- Mole fractions $X_i = \frac{N_i}{\sum_j N_j} = \frac{N_i}{N_{tot}}$
- Mass fractions $Y_i = \frac{M_i}{\sum_j M_j} = \frac{M_i}{M_{tot}}$

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Ex3 Constant Volume Explosion



Exercise 3, Constant Volume

combustion of 0.15 moles of acetylene (C_2H_2) and 2 moles of air. Volume of vessel is 50 liter.
 $T=800K$.

Initial State

given molar fraction of O_2 in air is 0.21 and N_2 in air $1-0.21=0.79$.

- Determine $X_{i,1}$ of initial mixture
- Determine $Y_{i,1}$ of initial mixture
- Determine $u_{mix,1}$ using the polynomials!

Final State

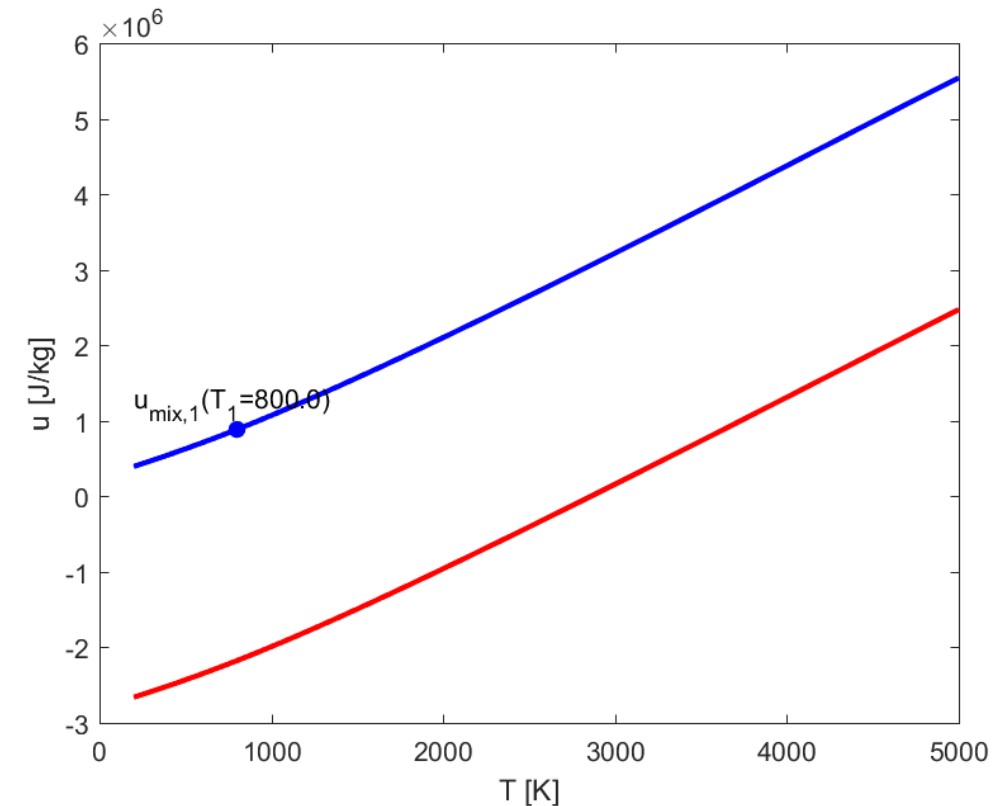
- Determine $X_{i,2}$ of final mixture (after combustion!)
- Determine $Y_{i,2}$ of final mixture
- Determine $u_{mix,2}(T)$ for a range of T and plot it. Compare to $u_{mix,1}$
- Determine T_2 and p_2 .

Ex3 Constant Volume Explosion

$$u_{mix,2} - u_{mix,1} = 0$$

- Exercise 3 Approach

- Determine X_i of mix
- Determine Y_i of mix
- Determine $u_{mix,1}$ using the polynomials!
- Determine $X_{i,2}$ of mix (nb after combustion)
- Determine $Y_{i,2}$ of mix
- Determine $u_{mix,2}(T)$ for a range of T . Compare to $u_{mix,1}$
- Determine T_2 and p_2 .



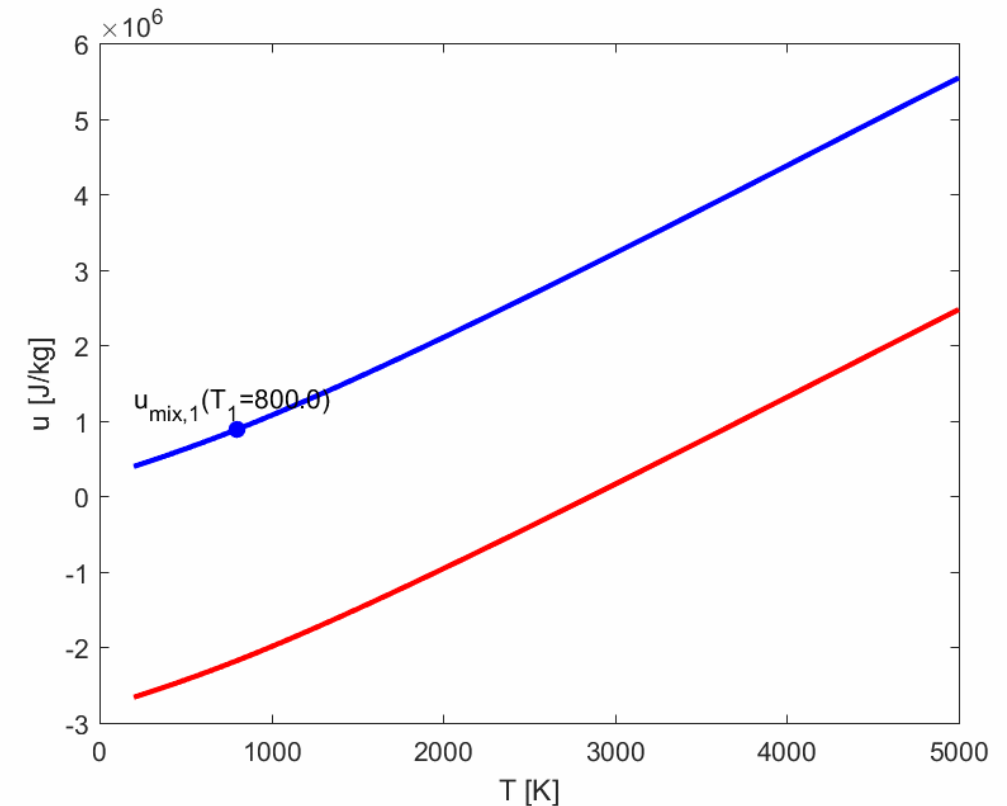
Ex3 Constant Volume Explosion

$$u_{mix,2} - u_{mix,1} = 0$$

• Exercise 3 Approach

- Determine X_i of mix
- Determine Y_i of mix
- Determine $u_{mix,1}$ using the polynomials!
- Determine $X_{i,2}$ of mix (nb after combustion)
- Determine $Y_{i,2}$ of mix
- Determine $u_{mix,2}(T)$ for a range of T . Compare to $u_{mix,1}$
- Determine T_2 and p_2 .

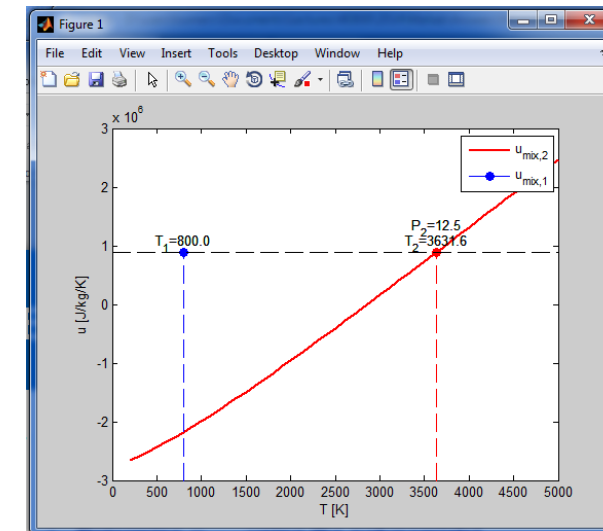
Note: A similar but **slightly different** approach is needed to compute the T rise in the combustor of the Jet Engine!



Ex3 Constant Volume Explosion

- **Exercise 3**
 - Determine X_i of mix
 - Determine Y_i of mix
 - Determine $u_{mix,1}$ using the polynomials!
 - Determine $X_{i,2}$ of mix (nb after combustion)
 - Determine $Y_{i,2}$ of mix
 - Determine $u_{mix,2}(T)$ for a range of T . Compare to $u_{mix,1}$
 - Determine T_2 and p_2 .

		X_i	Y_i			X_i	Y_i		
		Initial				Final			
C2H2		0.069767	0.063398		0.000000	0.000000			
O2		0.195349	0.218151		0.021687	0.023373			
N2		0.734884	0.718451		0.761446	0.718451			
CO2		0.000000	0.000000		0.144578	0.214312			
H2O		0.000000	0.000000		0.072289	0.043864			
		Initial				Final			
T	[K]	800.00			3631.62				
P	[bar]	2.860178			12.530943				



Contents

- **Ideal Gas mixtures**
 - **Thermodynamic Properties of a mixture (2.9)**
 - **Where hides combustion**
 - **Adiabatic flame temperature, an application of Nasa thermodynamic tables.**
- **Cycle analysis of a jet engine using thermodynamic tables**
 - **Combustor**
 - **Diffusor**
 - **Etc..**